

SEMESTER-III

UMT305: APPLIED THERMAL AND FLUID ENGINEERING				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: This course introduces basic concepts of thermodynamics, various thermodynamic cycles, I.C. engines and their performance parameters. This course also introduces fundamentals pertinent to fluid flow.				
Syllabus Basic Thermodynamics: Thermodynamics system & properties, Various processes, Energy and its form. First Law of Thermodynamics, Concept of internal energy & enthalpy, Second Law of Thermodynamics & its Corollaries, Kelvin Plank and Clausius statements, Reversible and Irreversible processes, Carnot cycle, Clausius theorem and concept of entropy, Principle of increase of entropy. Vapor Power Cycles: Properties of pure substance, Thermodynamic diagrams, Steam tables and their use, Moiller Diagram, Rankine cycle and modified Rankine cycle, reheat, regenerative cycles. Air Standard Cycles and I. C. Engines: Introduction and assumptions of air standard cycles, Otto cycle, Diesel cycle and Dual cycle. Comparison of Otto, Diesel and Dual cycles. Classification and application of I.C. engines, Combustion in I.C. engines, Flame propagation, pre-ignition, detonation, mixture requirements, fuel rating, Performance of I.C. engines. Introduction to Fluid Mechanics: Definition of a fluid and its properties, Hydrostatics, Measurement of pressure and thrust on submerged surfaces. Principles of Fluid Motion: Description of fluid flow, Continuity equation, Euler and Bernoulli equations; Pitot total head and static tubes, Venturi-meter, Orifice-meter, Rotameter, Momentum equation and its applications. Pipe Flow: Fully developed flow, Laminar pipe flow, Turbulent pipe flow, Major and minor losses, Hydraulic gradient line (HGL) and total energy line (TEL).				
Laboratory Work 1. Study of performance parameters of Dual fuel engine using syngas from waste materials-sustainable energy efforts 2. Study of performance parameters of Diesel engine 3. Study of performance parameters of VCR engine 4. Verification of Bernoulli's theorem 5. Determination of hydrostatic force and its location on a vertically immersed surface 6. Determination of friction factor for pipes of different materials 7. Determination of loss coefficients for various pipe fittings 8. Visualization of laminar and turbulent flow				
Course Learning Outcomes (CLOs) The students will be able to: 1. apply the first and second laws of thermodynamics for the thermal analysis of vapor power cycle and air standard cycles. 2. determine and analyze the performance parameters of I.C. engines. 3. analyze and solve problems of fluid-based engineering systems.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

4. evaluate practical problems associated with pipe flow systems.

Text Books

1. Thermodynamics: An Engineering Approach, Cengel, Y. A. and Boles, M., Tata McGraw Hill, 2008.
2. Internal combustion engines, Ganesan, V., McGraw Hill Education (India) Pvt Ltd., 2012
3. Fluid Mechanics, Cengel, Y. A., Tata McGraw Hill Education, 2010
4. Fluid Mechanics and Fluid Power Engineering, Kumar, D. S., S. K. Kataria, 2009

Reference Books

1. I.C .Engine, Heywoold, J. B., McGraw Hill, New Delhi, 1988.
2. Basic Engineering Thermodynamics, Joel, R., Pearson Education Asia, New Delhi, 1996.
3. Thermodynamics & Heat Power, Granet, I., Pearson Education Asia, New Delhi, 2003.
4. Power Plant Engineering, Nag, P. K., Tata McGraw Hill, New Delhi, 2008.

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	35

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